

1. Administrative & Study Identification

Full Study Title: A multicenter, open-label Phase 3 study: ambulatory blood pressure monitoring in adult patients with chronic spontaneous urticaria inadequately controlled by H1-antihistamines treated with remibrutinib up to 12 weeks **Short Title/Acronym:** Remibrutinib ABPM Study in CSU **Protocol Number:** CLOU064A2305 **NCT Number:** NCT05795153 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Remibrutinib (LOU064) **Phase of Development:** Phase 3 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To rule out an increase of >3 mmHg in 24-hour average Systolic Blood Pressure (SBP) at steady state (Week 4) compared to baseline, as assessed by Ambulatory Blood Pressure Monitoring (ABPM). **Secondary Objectives:** To assess overall safety, tolerability, and efficacy over 12 weeks in adult participants with Chronic Spontaneous Urticaria (CSU) inadequately controlled with second-generation H1-antihistamines (H1-AH).

2.2 Background & Rationale Chronic Spontaneous Urticaria (CSU) is a mast cell-mediated skin disease presenting with wheals and angioedema. Many patients remain inadequately controlled on first-line H1-antihistamines. Remibrutinib is a novel, oral Bruton’s tyrosine kinase (BTK) inhibitor being developed for CSU. To support its long-term chronic use profile, this study specifically assesses the cardiovascular pressor effects of remibrutinib using continuous ABPM, in accordance with FDA guidance for drugs intended for chronic use.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Single Group Assignment) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 144 evaluable patients **Number of Sites:** 44 global locations **Estimated Study Start:** 05-Apr-2023 **Estimated Primary Completion:** 25-Apr-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female adult participants \$\\ge 18\$ years of age. 2. CSU duration for \$\\ge 6\$ months prior to screening, inadequately controlled by second-generation H1-AH. 3. Documentation of hives within three months before baseline. 4. Willing and able to complete an Urticaria Patient Daily Diary (UPDD) without missing more than 2 entries in the 7 days prior to baseline.

4.2 Exclusion Criteria 1. Participants unable to tolerate 24-hour ambulatory blood pressure measurement using an automatic ABPM device. 2. Ongoing or past history of hypertension and/or SBP \$\\ge 140\$ or \$\\le 90\$ mmHg OR DBP \$\\ge 90\$ or \$\\le 60\$ mmHg at screening. 3. Participants taking or requiring medications prohibited by the protocol (including drugs known to interfere with blood pressure assessments). 4. Participants working night shifts.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Remibrutinib 25 mg twice a day (b.i.d.), to be taken morning and evening with a 12-hour interval. Route of Administration: Oral (film-coated tablet). Duration of Treatment: 12 weeks of open-label active treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Background maintenance therapy with second-generation H1-antihistamines. Prohibited: Use of any antihypertensive medications or other concomitant drugs known to interfere with blood pressure evaluations.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Up to -28 Days | Informed Consent, Medical History, Baseline BP Eligibility Check | | **Baseline** | Day 1 | 24-hour ABPM initialization, First Dose, Urticaria Diary Review | | **Interim** | Week 4 | Primary Endpoint: 24-hour ABPM Evaluation, Safety Assessment | | **End of Treatment** | Week 12 | Final Efficacy & Safety Evaluation, IP Accountability | | **Follow-Up** | Week 16 | 4-week treatment-free safety follow-up |

(Note: Following the 12-week period, eligible patients had the option to enter a long-term extension study, CLOU064A2303B).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in 24-hour weighted average Systolic Blood Pressure (SBP) from baseline to Week 4, designed to rule out a mean increase ≥ 3 mmHg. **Measurement Method:** 24-hour continuous Ambulatory Blood Pressure Monitoring (ABPM).

7.2 Secondary & Exploratory Endpoints * Overall safety and tolerability over 12 weeks (Treatment-Emergent Adverse Events, laboratory derivations). * Overall efficacy in CSU symptom control over 12 weeks.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Safety evaluations collected via patient diaries, vital signs, ECGs, and central lab tests measuring TEAEs. **Serious Adverse Events (SAEs):** Routine SAE monitoring per standard 24-hour regulatory timelines. **Data Monitoring Committee (DMC):** Internal safety review boards governing liver enzyme abnormalities and specific renal alert criteria monitoring.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients who received at least one dose of remibrutinib (Safety Set). **Sample Size Justification:** Powered to rule out a true difference of ≥ 3 mmHg in SBP at steady state (Week 4). **Handling of Missing Data:** The Mixed Models for Repeated Measures (MMRM) approach was utilized. If participants took prohibited antihypertensive treatment before Week 4, subsequent measurements were excluded. Excluded measurements were imputed by artificially increasing the 24-hour SBP by 3 mmHg from the baseline value.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard site visits to verify Urticaria Patient Daily Diary (UPDD) completion and proper ABPM operational deployment. **Audit Trail:** eCRF entry tracking, specifically for ABPM central data extraction and TEAE clinical laboratory derivation flagging.

11. Study Identifiers & Governance

Primary ID: CLOU064A2305 **Registry Number:** NCT05795153 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** LOU064/Remibrutinib ABPM Study in CSU **Official Regulatory Title:** A multicenter, open-label Phase 3 study: ambulatory blood pressure monitoring in adult patients with chronic spontaneous urticaria inadequately controlled by H1-antihistamines treated with remibrutinib up to 12 weeks

12. Clinical Framework (CSU Specific)

Primary Condition: Chronic Spontaneous Urticaria (CSU) **Diagnosis Threshold:** CSU duration for ≥ 6 months, inadequately controlled by H1-AH **Medical Methodology:** BTK Inhibition combined with continuous cardiovascular telemetry **Optimization Target:** SBP increase ≤ 3 mmHg and optimal CSU symptom reduction

13. Study Procedures & Methodology

Management Pathway: BTK Inhibitor (Remibrutinib) Step-up Therapy **Education Protocol (HCP):** 24-hour automatic ABPM device application, calibration, and patient instructions **Education Protocol (Patient):** Strict Urticaria Patient Daily Diary (UPDD) logging (morning and evening inputs) **Quality Audit Mechanism:** Assessment of diary compliance and automated verification of baseline and Week 4 ABPM successful data capture

14. Observation & Follow-Up Schedule

Total Duration: 20 Weeks (4W screening + 12W treatment + 4W follow-up) **Onsite Visit Cadence:** Baseline, Week 4, Week 12, Follow-up (Week 16) **Remote Monitoring Frequency:** Twice-daily patient diary inputs **Checklist Review Interval:** Routine site verification to ensure no more than 2 missing UPDD entries over trailing 7-day epochs

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				